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Effect of work duration on physiological and rating scale of perceived exertion responses during self-paced interval training

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This study compared running velocity, physiological responses, and perceived exertion during self-paced interval training bouts differing only in work bout duration. Twelve well-trained runners (nine males, three females, 28 ± 5 years, $VO_{2\,max}$ 65 ± 6 mL min^{-1} kg^{-1}) performed preliminary testing followed by four "high-intensity" interval sessions (Latin squares, 1 session week^{-1} over 4 weeks) consisting of 24 × 1, 12 × 2, 6 × 4, or 4 × 6-min running bouts with a 1:1 work-to-rest interval (total session duration 48 min). The average running velocity decreased (93%, 88%, 86%, 84% $v$$VO_{2\,max}$, $P < 0.01$) with increasing work duration. Peak VO_2 averaged about 92 ± 4% of $VO_{2\,max}$ for 2-, 4-, and 6-min intervals compared with only 82 ± 5% for 1-min bouts ($P < 0.001$). Six of 12 athletes achieved their highest average VO_2 and heart rate during 4-min intervals. The average RPE_{peak} (rating scale of perceived exertion) was ~17 ± 1 for all four interval sessions. RPE increased by 2–4 U during an interval training session. The mean lactate concentration was similar across sessions (4.3 ± 1.1–4.6 ± 1.5 mmol L^{-1}). Under self-paced conditions, well-trained runners perform "high-intensity" intervals at an RPE of ~17, independent of interval duration. The optimal interval duration for eliciting a high physiological load is 3–5 min under these training conditions. Increases in RPE during an interval bout are not associated with increasing blood lactate concentration.

The assumed advantage of interval training, organizing exercise as intermittent bouts of higher and lower intensity rather than one continuous bout, is the accumulation of a greater training stimulus at high exercise intensities (Åstrand et al., 1960; Christensen et al., 1960). *Interval training* has come to encompass a broad array of training prescriptions. However, for a given exercise modality, any interval training prescription consists of five variables: (1) work interval duration and (2) intensity, (3) recovery interval duration and (4) intensity, and (5) total work duration (interval number × interval duration). These variables can be manipulated to generate a potentially endless number of specific interval training session prescriptions (Thibault & Marion, 1999). *Aerobic* interval training for the purpose of enhancing maximal aerobic capacity and endurance performance typically involves a work interval duration ranging from ~1 to 8 min and a rest interval varying from 30 s to 5 min (Fox et al., 1975). A work-to-rest interval between 2:1 and 1:2 is common, although work-to-rest ratios of up to 5:1 are used. The prescribed exercise intensity for intervals in this range is generally 85–105% of maximal aerobic power (Thibault & Marion, 1999).

Little empirical data is available quantifying how variation in the variables of the interval training prescription impacts the athlete's *training response*, the achieved intensity, physiological responses, and perceived exertion elicited. Published data quantifying acute physiological responses associated with different aerobic interval training conditions are limited and almost entirely based on the measurement of physiological responses to fixed work intensities at a predefined percentage of maximal aerobic power or velocity (Patterson et al., 1997; Billat et al., 1999, 2000a, b; Demarie et al., 2000; Stepto et al., 2001).

In the normal training setting, precise external control of intensity is unusual. In many sports, work intensity is not a constant function of velocity due to variable terrain, wind, water conditions, snow conditions, etc. Coaches or athletes typically prescribe interval training sessions using manipulation of interval duration (distance or time), rest duration, and number of work bouts. The actual exercise intensity achieved can be defined as a dependent variable constrained by several independent variables: goal intensity, work duration, recovery time, total volume of work, and the perception of exertion and response to that perception. The purpose of this study was therefore to quantify how work interval duration impacts physiological responses and perceived exertion during training sessions where well-trained endurance athletes are uniformly instructed to perform a "high-intensity interval session".

Methods

Subjects and preliminary testing

Twelve athletes from a local running club (nine males, three females) volunteered to participate in this investigation, which was approved by the human subjects research review board of the Department of Health and Sport, Agder University College. Criteria for participation in the study were: (1) regular training, (2) competition in running races, and (3) regular use of interval type training sessions. Participants were informed of the risks associated with the study and their choice to terminate participation at any time. They had been training for running events, either track and field or orienteering, for ≥ 4 years, and averaged two interval-type training sessions per week out of an average of 5.4 total training sessions per week (range 4–9) in their current training. All subjects were comfortable running at high speeds on a motorized treadmill. Throughout the study, subjects maintained their pre-study volume of training.

Initially, a continuous ramp protocol run to exhaustion was performed to quantify ventilatory thresholds (VT_1 and VT_2), maximal heart rate (HR_{max}), maximal oxygen consumption ($VO_{2\,max}$), running velocity at maximal oxygen consumption ($vVO_{2\,max}$), and peak blood lactate concentration ($[La^-_{peak}]$). The maximal treadmill test was repeated following the data collection period to control for possible significant changes in physiological capacity. Both testing and subsequent interval training bouts were performed on a motorized treadmill (Woodway P55 Sport, Weil am Rhein, Germany) at 1% elevation (Jones & Doust, 1996). After a 20–30-min warm-up, the maximal treadmill test was initiated with a 2-min run at $9\,km\,h^{-1}$ ($150\,m\,min^{-1}$) with subsequent increases of $0.4\,km\,h^{-1}$ ($6.7\,m\,min^{-1}$) every 30 s until voluntary exhaustion.

Gas exchange measurements

Expired gas samples were measured continuously using a Cortex Metamax gas analyser (Cortex Biophysik GmbH, Liepzig, Germany) and a Triple-V turbine (Sensormedics BV, Bilthoven, The Netherlands). Mixing chamber derived gas exchange data were averaged over 10-s time intervals. Calibration was performed prior to each test according to the manufacturer's instructions. This system has been previously shown to be valid, and highly stable over up to 120 min of continuous measurement (Schulz et al., 1997). HR responses were collected every 15 s using a telemetry system (Polar Electro, Kempele, Finland). $VO_{2\,max}$ was defined as the highest 30-s average measurement recorded during the ramp test. The running velocity at maximal oxygen consumption ($vVO_{2\,max}$) was defined as the lowest treadmill velocity corresponding to the 30-s period defining $VO_{2\,max}$. VT_1 was defined as an increase in V_E/VO_2 without an increase in V_E/VCO_2. VT_2 was defined as the point where V_E/VCO_2 also started to rise.

Blood lactate measurements

At 1 and 3 min after voluntary exhaustion during the maximal treadmill test, venous blood was sampled via finger stick to identify peak lactate concentration. In all subjects, lactate values at 3 min were equal to or already below the 1-min values. Twenty microlitres capillary blood samples were immediately analysed for whole, non-haemolysed blood lactate concentration using a YSI 1500 Sport lactate analyser calibrated with reference standards before and after each test. Values were subsequently corrected for the ~20% of total blood lactate trapped in red blood cells using a multiplication factor of 1.25 (Medbø et al., 2000).

Interval training bouts

During the 4 consecutive weeks following preliminary testing, each athlete replaced one regularly scheduled hard training session with an interval bout performed in the laboratory. The four interval workouts varied in interval duration: 1-, 2-, 4-, or 6-min work periods. The work-to-rest ratio was fixed at 1:1 and the total work at 24 min for each session (i.e., 24×1, 12×2, 6×4, and 4×6 min). The order of interval bout exposure was randomized using a Latin squares design. Each subject performed the four interval sessions at the same time of day each week. Subjects were instructed to treat each weekly training session as a "high-intensity interval session". They were also instructed to attempt to maintain the highest average running velocity they could across all the work bouts of each interval session.

The end of the 20–30-min warm-up was used to determine the starting velocity for the first interval. Thereafter, the treadmill velocity could be increased or decreased at any time via a hand signal to the test administrator controlling the treadmill via PC. At regular, frequent intervals during each work bout, subjects were verbally encouraged and asked whether they desired "more or less speed". Between work bouts, subjects were also free to select their own recovery intensity. About 15 s prior to each new work bout, subjects straddled the treadmill belt as it was accelerated up to the athlete's desired starting velocity, using the velocity of the previous interval as a reference. At the conclusion of a running bout, athletes held the handrails and hopped to the sides of the treadmill surface as the velocity was rapidly reduced. They were then free to choose their activity intensity during the recovery period. Two small, motorized fans, positioned at each side and chest high, were used to ensure effective evaporative cooling. The laboratory temperature during the training sessions was 18–21 °C. Treadmill calibration demonstrated that treadmill velocity was highly accurate over a broad range of velocities.

Physiological measurements during interval training

Running velocity, gas exchange data, and heart rate were collected continuously during each interval training session. The total workout duration varied slightly (42–47 min) across the four conditions, since the last time period was a rest interval of varying length. Two blood samples were collected during each interval training session: 60 s after the 12th minute of interval training, and 60 s after the completion of the last work interval.

Perceived exertion measurements

The 15-point Borg rating scale of perceived exertion (RPE) (Borg, 1970) was used to quantify global perceived exertion during each interval session. Subjects were provided with standard, written instructions translated into Norwegian. RPE was determined during the final 10 s of every other interval bout for the 1- and 2-min interval sessions and each of the 4- and 6-min interval bouts.

Training control

This study was carried out during the early pre-competition preparation phase of training (February–early March). Training control was confirmed via a training diary. Subjects were instructed to abstain from hard training the day prior to laboratory sessions. No special attempt was made to control the diet of the athletes; they were merely reminded to come to the training session well hydrated and non-fasted.

Statistical analyses

Physiological and RPE responses during the four different interval sessions were compared using the General Linear Model with a repeated measures design (SPSS 10.0, Chicago, Illinois, USA). An α level of 0.05, after Bonferroni adjustment to control for the increased risk of Type I error associated with multiple comparisons, was considered to be statistically significant. Where appropriate, pairwise comparisons of physiological data were made using the paired samples *t*-test with the α level set at 0.05.

Results

Maximal treadmill testing

Subject characteristics and results from preliminary testing are presented in Table 1. The average $VO_{2\,max}$ of the three females in the study was 60 and 67 mL^{-} min^{-1} kg^{-1} among the nine males, suggesting that the males and females were of similar relative capacity. The combination of $VO_{2\,max}$, ventilatory threshold determinations, and self-report of current training confirmed that the subjects were all representative of a regularly training, non-elite athlete population. The intermittent treadmill test to exhaustion was repeated following the period of interval training data collection, which lasted 4–5 weeks. The treadmill distance at exhaustion (3456 ± 441 vs. 3631 ± 543 m) improved by 5% ($P<0.001$) in the follow-up test. HR_{max} also declined significantly (190 ± 10 vs. 187 ± 10 beats min^{-1} (bpm), $P<0{,}05$). However, $VO_{2\,max}$ (65.3 ± 5 vs. 65.6 ± 6 mL min^{-1} kg^{-1}), ventilatory thresholds, and $[La^{-}]_{peak}$ (8.4 ± 2 vs. 8.6 ± 2 mmol L^{-1}) did not change significantly over the course of the study. The relative intensity of the interval training sessions was therefore calculated based on the results of the preliminary treadmill test.

Physiological responses during interval sessions

Running velocity

Figure 1 depicts the average running velocity for each intermittent bout of all four interval sessions. An inspection of the individual data showed that three of the 12 athletes underestimated their sustainable velocity during the initial bouts and increased during the following bouts (as much as 12%), while the remainder appeared to target their average velocity more accurately from the start. When the average velocity of the first 4 min and the last 4 min of each session were compared, the upward adjustment in velocity averaged 5%, 4%, 4%, and 1.5% for 1-, 2-, 4-, and 6-min work durations, respectively, and was statistically significant for the 1-, 2-, and 4-min interval sessions ($P<0.005$). Figure 1 also reveals that the average running velocity declined with each increase in work bout duration, from $93 \pm 2\%$ $vVO_{2\,max}$ for the 24×1-min interval session to $83 \pm 2.5\%$ during the 4×6-min interval session ($P<0.005$ for each duration comparison). Expressed as a percentage of the velocity corresponding to the second ventilatory threshold (vVT_2), the interval sessions were performed at $113 \pm 7\%$, $107 \pm 5\%$, $105 \pm 6\%$, and $102 \pm 4\%$ of vVT_2 for the 1-, 2-, 4-, and 6-min work durations, respectively.

VO_2 and heart rate responses

When athletes self-selected their running velocity, the peak oxygen consumption (92–93% of $VO_{2\,max}$) and oxygen consumption during recovery (26–31% $VO_{2\,max}$) were not statistically different for work durations of 2, 4, or 6 min (Table 2). Individual responses to the four training prescriptions are presented in Fig. 2. Six of 12 subjects achieved their highest oxygen consumption when performing 4-min work intervals. Peak VO_2 during 1-min intervals was significantly lower ($82 \pm 5\%$ $VO_{2\,max}$ vs. 92–93% $VO_{2\,max}$ for 2-, 4-, and 6-min intervals $P<0.001$), and recovery oxygen consumption significantly higher ($46 \pm 5\%$ $VO_{2\,max}$ vs. 26–31%, $P<0.001$) compared with the 2-, 4-, and 6-min work interval duration sessions. However, the average oxygen consumption

Table 1. Characteristics of the subjects ($N=12$, nine males, three females)

Age (years)	Height (cm) Weight (kg)	$VO_{2\,max}$ (ml kg^{-1} min^{-1})	HF_{max} (beats min^{-1})	VT_1 (%$VO_{2\,max}$)	VT_2 (%$VO_{2\,max}$)	$vVO_{2\,max}$ (km h^{-1})
28 ± 5	176 ± 8 68 ± 10	65 ± 6	190 ± 10	68 ± 6	84 ± 4	19.7 ± 1

$N=12$ values are presented as mean $\pm$ SD. The velocity at $VO_{2\,max}$ was determined at a constant incline of 1%.

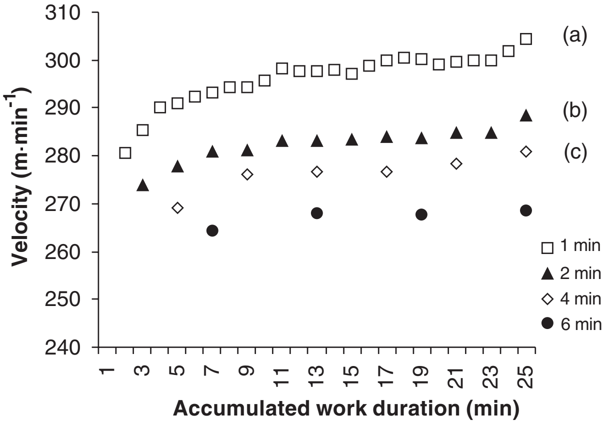

Fig. 1. Mean velocity (m min^{-1}) for each work bout performed during 1-, 2-, 4-, and 6-min work duration interval sessions. Error bars are omitted for clarity. The mean running velocity was significantly different for each work duration condition. (a, b, c): The initial running velocity was significantly lower than subsequent running bouts (>8 min accumulated work duration, $P<0.05$).

Table 2. Oxygen consumption responses to four interval bouts differing in work duration

Work bout (and recovery) duration (min)	Peak VO_2 ($\%VO_{2\,max}$)	VO_2 at end of recovery periods ($\%VO_{2\,max}$)	Average VO_2 for entire session ($\%VO_{2\,max}$)
1	81.8±5.2*	46.0±3.4*	61.3±3.9
2	92.4±4.4	27.5±5.3	60.7± 4.7
4	93.3±4.8	25.6±8.1	59.4±4.2
6	91.7±3.0	30.6±11.6	61.4±3.3

Values = mean±SD.
*$P<0.01$ vs. 2-, 4-, and 6-min work duration trials.

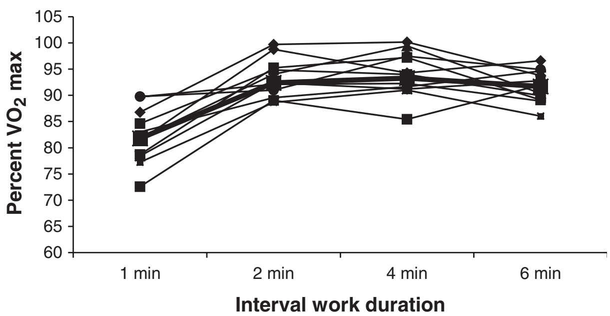

Fig. 2. Individual peak VO_2 during interval sessions with 1-, 2-, 4-, or 6-min work duration. The bold line indicates the averaged peak VO_2 response under each condition.

for the entire session (even numbers of work and rest periods) was virtually identical for all four conditions (59.4–61.4% $VO_{2\,max}$). Separate repeated measures analyses were performed to examine whether a testing order effect on exercise intensity was present. Whether measured as peak HR or peak VO_2, the order of testing did not significantly influence self-selected exercise intensity.

Figure 3 displays the averaged heart rate responses throughout the four interval training sessions. The peak heart rate (Fig. 4) was slightly but significantly lower during the 1-min interval, compared with the 4- and 6-min intervals (92% for 1-min intervals vs. 95% of HR_{max} for both 4- and 6-min intervals, $P<0.005$). Peak heart rate responses during the 2-min intervals were only significantly different from the 4-min interval responses ($P=0.05$). Overall, the differences in peak heart rate responses between the 1-min intervals and the longer interval durations were smaller than the differences in peak oxygen consumption.

Blood lactate concentration

Blood lactate concentration, measured immediately after the 12th and 24th minute of interval running during each interval session, was similar across interval duration and approximated 4.5 mM (Fig. 5). Blood lactate concentration tended to rise slightly from midway to the end of the interval sessions under all conditions. However, this increase was only statistically significant during 1-min intervals (5.0±1.4 vs. 3.9±1.1 mM, $P=0.02$).

Ratings of perceived exertion

RPE increased steadily throughout an interval session (Fig. 6) under all work duration conditions. The increasing perception of exertion during the interval sessions was well fit by a linear regression line (RPE = 14.5±0.11 × (accumulated work duration in minutes)). RPE tended to be lower at the end of the initial 1-min bouts compared with the 4- and 6-min interval bouts, consistent with the greater degree of velocity adjustment observed for the shortest interval duration. However, peak RPE was virtually identical under all four work-duration conditions (ranging from 16.8±1 to 17.2±1).

Discussion

The key finding of this study is that when prescribed interval training sessions vary in work period duration from 1 to 6 min, athletes adjust their intensity such that blood lactate and perceived exertion responses throughout each session are essentially identical. To our knowledge, this study is unique in that subjects self-selected their exercise intensity throughout each session in response to a standardized work prescription, conditions that are similar to how endurance athletes normally train.

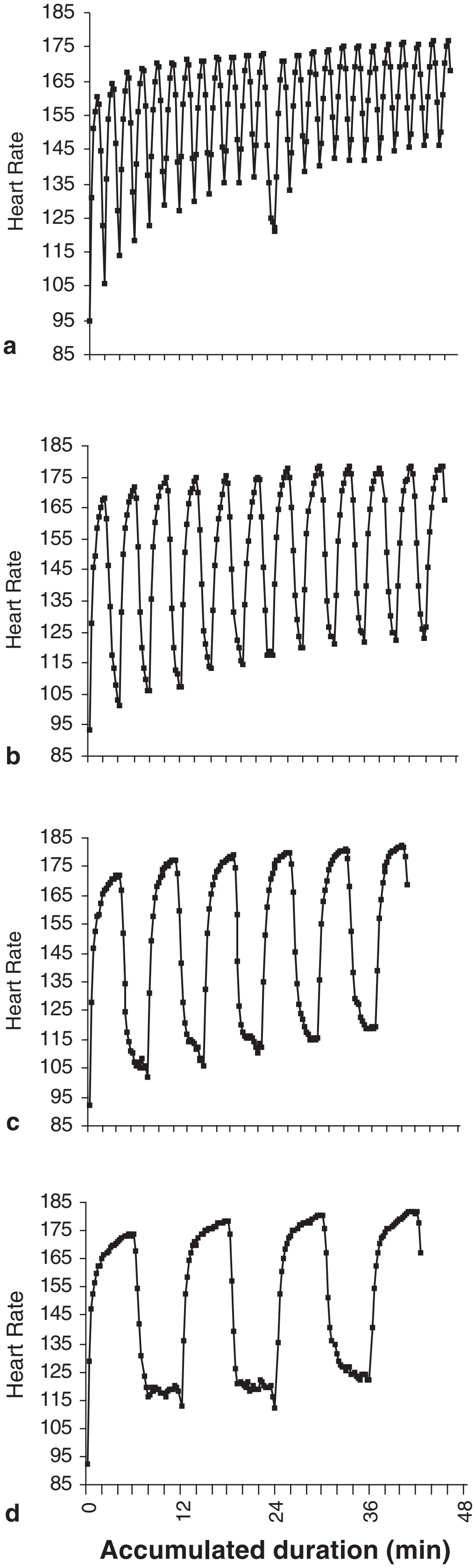

Fig. 3. Mean heart rate responses during (a) 1-, (b) 2-, (c) 4-, and (d) 6-min work bout duration conditions. Data presented are based on results from 10 of 12 subjects because continuous telemetry recordings were partially lost for two subjects.

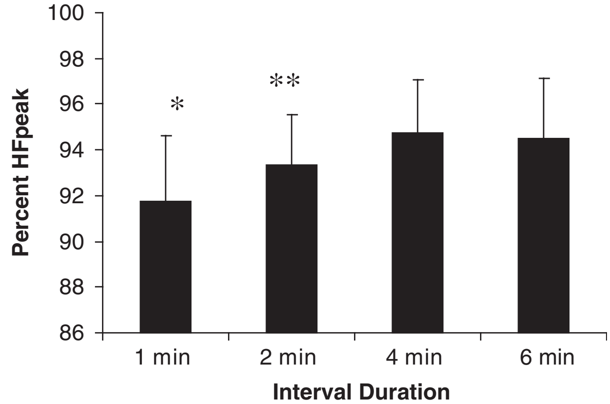

Fig. 4. Group means ($\pm$SD) for peak heart rate (% HF peak) during 1-, 2-, 4-, and 6-min work bout duration sessions. $^{*}P<0.05$ vs. 4- and 6-min conditions. $^{**}P = 0.05$ vs. 4-min condition.

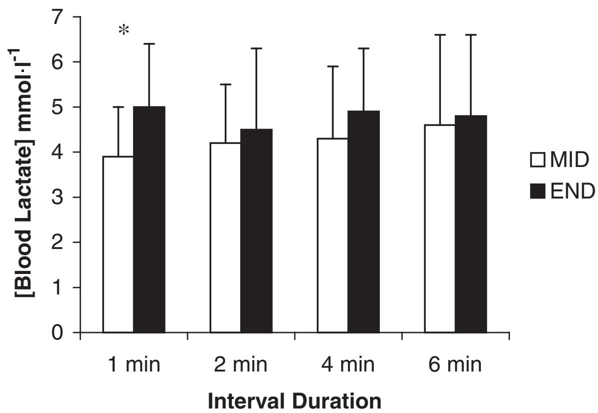

Fig. 5. Group means ($\pm$SD) for blood lactate measurements performed after 12 min (MID, open bars) and after 24 min (END, solid bars) of accumulated work duration under the four different interval training conditions. $^{*}P<0.05$ vs. END.

The results also suggest that under self-paced conditions with a typical 1:1 work-to-rest ratio, peak physiological responses are very similar for interval durations between 2 and 6 min and significantly higher than those obtained during an equal duration of work performed as 1-min work intervals.

A work duration of 4 min appears to approximate an optimal duration for achieving peak cardiovascular responses under self-paced conditions. Six of 12 athletes achieved their highest oxygen consumption during 4-min work intervals. Our findings are interesting in the light of those of Stepto et al. (1999), who compared the cycling performance enhancing effects of five different interval training programmes built around work bouts of 0.5-, 1-, 2-, 4-, or 8-min duration. Power output was controlled for the

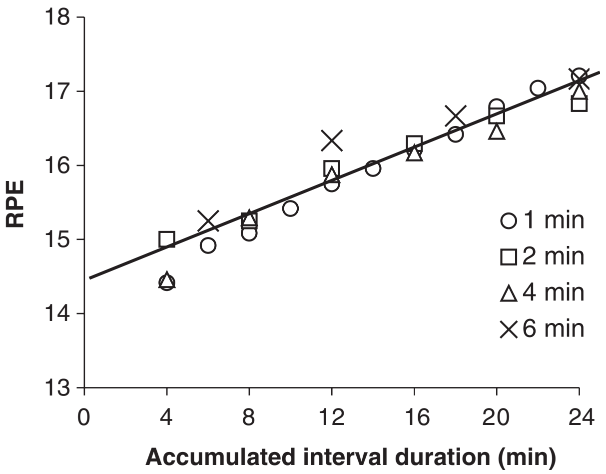

Fig. 6. Group means for rating scale of perceived exertion (RPE) recorded throughout the four different interval training sessions. Error bars are omitted for clarity. SD averaged $\sim$0.8 RPE units for all four conditions. RPE changes over time were well fit by a linear regression equation: RPE = 14.5+0.11 $\times$ accumulated work duration (min).

different interval training groups, and equal to 175%, 100%, 90%, 85%, and 80% of peak power output, respectively. Both peak sustained power output and 40-km time trial performance were improved most in the group of cyclists performing 4-min work bouts at 85% of PPO.

Another important aspect of the interval training description is the intensity of work during recovery periods. This intensity typically ranges from complete rest up to 50% of $VO_{2\,max}$ (Billat et al., 2000a). The intensity of work during rest can be expected to influence the rate of recovery of phosphocreatine, the rate of lactate elimination, and the oxygen kinetics at the onset of the next exercise bout. What strategy do athletes actually choose when prescribed a specific interval training task? In this study, the athletes appeared to choose recovery intensity sufficient to maintain oxygen consumption at 20–35% $VO_{2\,max}$. This intensity range is somewhat lower than that found to be optimal for lactate elimination after high-intensity work (McLellan & Skinner, 1982) and also lower than the $\sim$50% $VO_{2\,max}$, which has been suggested as optimal for achieving and maintaining the highest possible oxygen consumption levels during the work periods (Billat et al., 2000b). One-minute intervals actually resulted in lower VO_2 during the work portion and higher VO_2 during recovery, such that the "interval training" effect was truncated. In contrast, 4-min interval bouts were associated with the highest VO_2 during work and the lowest VO_2 during recovery (Table 2), although these differences were not statistically significant when compared with the 2- and 6-min work duration conditions. Very little research has been reported on the impact of recovery duration and/or recovery work intensity on the overall intensity maintained during interval training bouts. This is currently a topic of investigation in our laboratory.

As expected, heart rate responses tracked well with oxygen consumption curves. The peak heart rate was similar for the 4- and 6-min work bouts, equalling 95% of maximal heart rate. However, the difference in peak heart rate response ($\sim$91% HR_{max} vs. 95% HR_{max}) was smaller than the difference in peak oxygen consumption (82% vs. 93% $VO_{2\,max}$). In a training setting, heart rate may give a somewhat misleading picture of the relative oxygen consumption achieved during short intervals. One explanation for this discordance is the break from linearity in the heart rate-intensity relationship occurring around the second ventilatory threshold (Hoffmann et al., 1994, 1997). During preliminary testing, a clear negative heart rate inflection point was identified for 10 of the 12 athletes. This would result in a lower HR/VO_2 ratio at intensities above the inflection point.

The peak heart rate during each work bout increased over the course of an interval session. If we ignore the first interval bout that was often followed by velocity adjustments, peak heart rate drifted upwards $\sim$5 bpm over the course of an interval session. More striking however was the increase in end-recovery heart rate, or "recovery heart-rate drift" in an interval training context, observed over the course of the sessions (Fig. 2). Using the 2-min interval bouts as an example, we observed a $\sim$25 bpm increase in end- recovery heart rate from the end of the first work bout to the end of the next to last work bout. At the same time, end-recovery oxygen consumption remained stable (mean of first three bouts $1.17 \pm 0.34\,L\,min^{-1}$ vs. mean of last three bouts $1.19 \pm 0.34\,L\,min^{-1}$). The increase in end-recovery heart rate was greatest for the 1- and 2-min intervals, suggesting that primarily the fast phase of heart rate recovery was slowed over the course of the training session. Peak VO_2, recovery VO_2, and blood lactate concentration were stable over most of the session, suggesting attainment of a quasi-steady-state in and around the active muscle. This suggests that the fast phase of heart rate recovery is largely controlled by central mechanisms. If this is the case (see e.g., Savin et al., 1982), then using heart rate recovery as a guide for determining when to begin subsequent interval bouts would appear to have little direct connection to recovery processes at the muscular level.

Blood lactate responses were surprisingly constant for the four different interval training sessions. In isolation, the similarity in blood lactate responses suggests a similar overall contribution of anaerobic metabolism across the four different conditions, since averaged blood concentrations measured under the four work-duration conditions were not significantly different (4.2–4.5 mM). The data also suggest a diminishing reliance on lactate metabolism over the course of a given interval session, since blood lactate measurements changed little from midway to the end of the interval training sessions (Fig. 5).

The observation that most of the increase in blood lactate concentration occurs after the initial bouts of intermittent exercise is not novel. Stepto and colleagues measured physiological responses to repeat 5-min cycling bouts at 82.5% of the maximal aerobic power separated by 1-min recovery periods. In that study, blood lactate concentration increased to 4.7 mM after the first bout and remained between 5 and 6 mM over the remaining seven bouts of exercise. Oxygen consumption was 5–7% lower in that initial bout compared with the third, fifth, and seventh bouts (only these were reported), suggesting that the initially greater oxygen deficit accounts for the early rise in blood lactate, and that subsequent work was performed with a closer balance between lactate production and elimination, in part due to reduced reliance on glycolytic metabolism (Stepto et al., 2001). In the present study, we also observed nearly constant blood lactate concentration from the middle of the session to the end, suggesting that lactate production and elimination were in balance despite running velocities averaging 102–113% of $v\mathrm{VT}_2$. This interpretation is made with caution, however. The concentration of a substance in a fluid compartment is by definition the amount of substance divided by the fluid volume in the compartment being measured. Direct comparisons of blood lactate concentrations at the beginning, middle, and end of an interval session are complicated by the fact that the distribution volume for lactate will progressively increase over a period of 20–40 min, since lactate ions distribute slowly in the extracellular space (Medbø & Toska, 2001). Over the course of a short, intense interval training workout, blood lactate concentration could theoretically stabilize despite a continued mismatch between production and elimination.

Perhaps the most common measure of exercise intensity in the normal endurance training setting remains perception of effort. Across a spectrum of endurance sports, the power output for a given velocity changes with weather, wind, hill elevation, snow conditions, etc. The heart rate at a given intensity drifts over time, changes with different environmental conditions, and lags behind during short high-intensity exercise such as interval training. Therefore, the athlete's perception of effort remains a critical tool for matching their actual work intensity with session goals. Our results suggest that over a range of interval training prescriptions, endurance athletes are able to calibrate their exercise intensity accurately so that perceived exertion is held constant. We found that at an average lactate of about 4.5 mM, peak RPE was ~ 17 across all four interval conditions. Our data are consistent with those of Seip et al. (1991), who reported an average RPE of 16.5 in trained runners running at a fixed blood lactate concentration of 4.0 mM. There are few data available describing RPE responses during interval-type training sessions. Among our subjects, the perception of effort increased linearly with time over the course of all four interval training sessions (Fig. 5). This was true even when running velocity, ventilation, and blood lactate remained constant. It appears reasonable to recommend that a typical high-intensity "aerobic" interval session should feel "hard" (RPE 15–16) initially, and will be perceived as "very hard" (RPE 17–18) by the end of the workout.

Perspectives

In recent years, research has increasingly focused on the physiological responses to intermittent exercise. However, to our knowledge, this study is the first to compare the physiological and perceptual responses of athletes during different high-intensity *aerobic* interval training prescriptions under the free-range conditions typical of endurance training. We conclude that within a reasonable range of prescriptions typical of "aerobic" interval training, trained endurance athletes automatically manipulate work intensity appropriately to maintain similar levels of peak exertion (an RPE of about 17) and blood lactate. Blood lactate measurements suggest that a quasi-steady-state is achieved, while RPE climbs linearly at a rate that suggests an appropriate upper limit for total work bout duration of ~ 30 min. Four minutes may represent a median optimal work duration for eliciting a high cardiovascular stress to self-paced $\mathrm{VO}_{2\,max}$ intervals when the work-to-rest ratio is 1:1. Future studies aim to investigate the physiological and perceptual impact of manipulating the recovery duration component of the interval training prescription.

Key words: endurance, athletes, intermittent exercise, heart rate, maximal oxygen consumption, perceived exertion.

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